Home Security for Vulnerable Groups

Domain: Criminal Behaviour
Sub-theme: Housebreaking / Burglary against Vulnerable Groups

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Secondary Research Overview

- Data Source: Kaggle (2020–2024) containing 1980 burglary cases.
- Filtered 1438 cases where victims belonged to vulnerable groups (women, children, elderly).

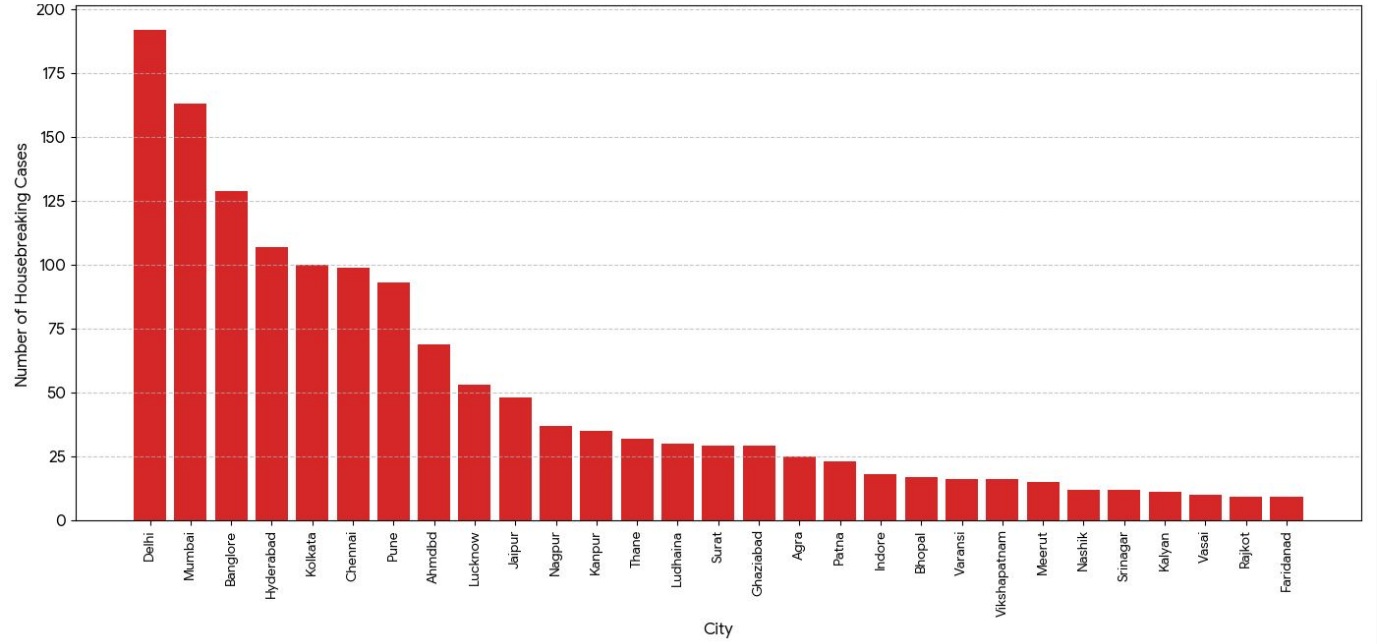
Supplementary Reference:

- NCRB data showing a steady rise in crimes against elderly and women in metro cities.
- HelpAge India report: 40% of India’s 112 million elderly live alone, increasing their vulnerability.
- OnlyIAS (by Physics Wallah) highlighted that weak physical & mental strength and declining joint families make elderly easy targets.

Key Insights from Secondary Research

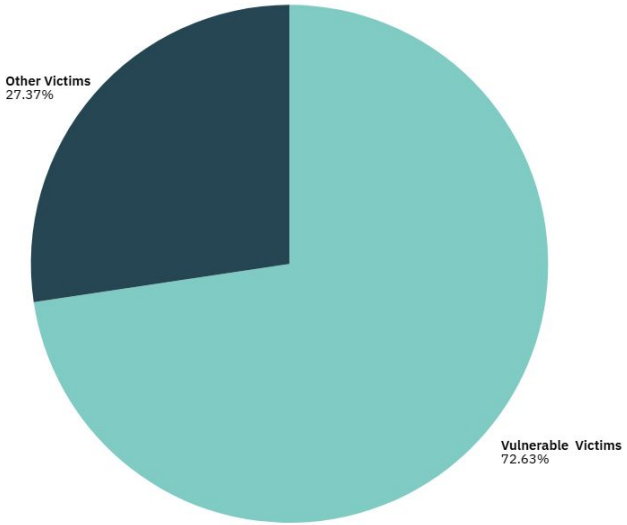
- Burglary cases are higher in urban and semi-urban regions.
- Elderly and women living alone are targeted more often.
- Offenders usually commit burglary for financial gain, or due to drug/loan pressure.
- The majority of burglaries take place during day hours when homes are unoccupied.
- Increased isolation, predictable routines, and poor security systems are key risk factors.

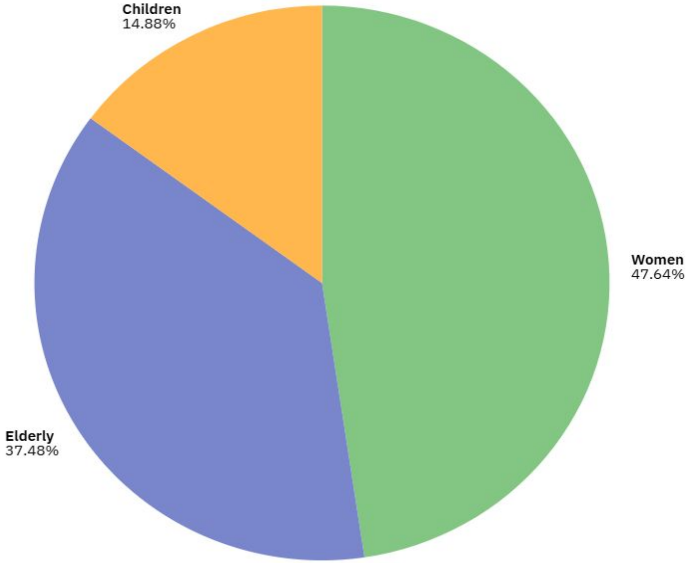
City-wise Distribution of Burglary Cases (2020–2024)



- Delhi and Mumbai reported higher burglary cases compared to Tier-2 cities.
- These cities also had greater incidents involving women and elderly victims.
- Urban anonymity & weak neighbourhood connections increase risk.

House Breaking cases reported 2020 - 2024





Observations:

- Around 73% of burglary victims in the vulnerable category were women and elderly.
- Children-related cases were lower but often involved theft during absence of adults.
- Elderly victims are often targeted in daylight hours, women mostly in early evening.

Weapons & Methods Used

- Most burglars entered through forced entry (doors/windows).
- Common tools used: knife, firearms, blunt objects.
- In only 14% of cases, no weapon was used.
- Some incidents involved deceptive entries like pretending to be service staff or delivery agents.

Primary Research Insights

To understand real-world experiences of burglary and perceived safety, brief interviews were conducted with police personnel, women, and elderly residents from local neighbourhoods.

Police Insights

> “Most burglary cases occur when the house appears unoccupied or predictable routines are noticed.”
> “Elderly and women living alone are often targeted due to limited physical resistance and easy trust.”

Key Takeaways:

- Criminals don’t follow a fixed schedule they choose the time based on their comfort.
- Lack of neighbourhood communication makes detection harder.
- Police emphasize awareness drives and CCTV visibility as deterrents.
- Most intrusions happen without use of advanced weapons — simple forced entry or bluffing as delivery staff.

Primary Research Insights

Women (Living Alone or Often at Home Alone)

> “Delivery people and maintenance workers make me slightly uneasy—especially when I’m alone.”
> “I usually check from the window or ask the guard, but not always.”

Key Takeaways:

- Verification habits vary — few use peepholes or cameras consistently.
- Women rely more on neighbours than police for immediate safety.
- Safety concerns peak during daytime deliveries or late-night knocks.
- CCTV presence provides emotional reassurance, even if rarely reviewed.

Primary Research Insights

Elderly (Staying Alone or Semi-Alone)

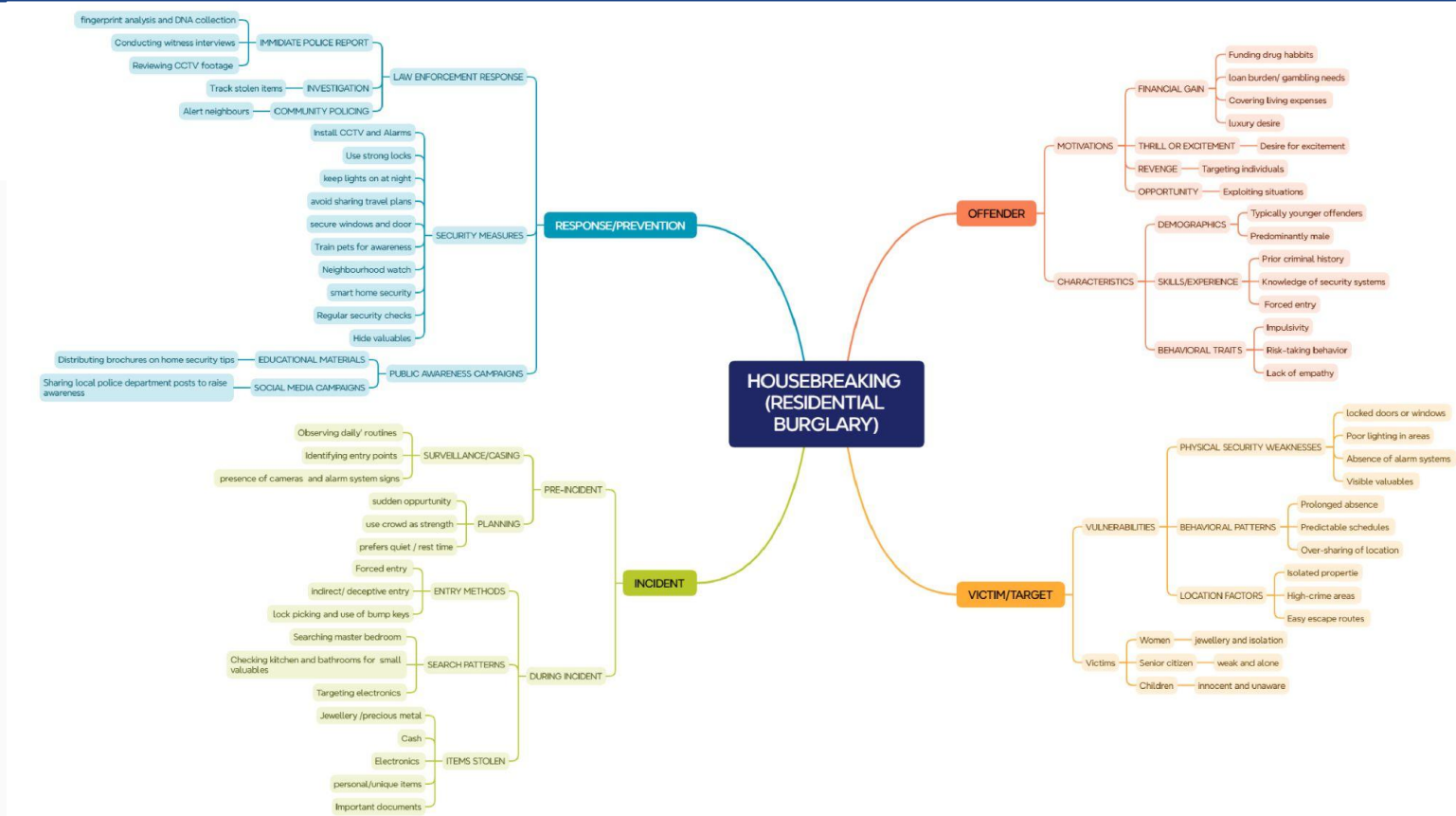
"I do feel scared when someone rings the bell and I can't recognize them."

"I don't fully understand how to use the door camera my son installed."

Key Takeaways:

- High fear and hesitation while answering doors.
- Low confidence with technology (CCTV, smart locks).
- Dependence on guards or neighbours for validation.
- Believe police respond slower in elderly cases.
- Feel targeted due to trusting nature and isolation.

Mind Map: System Understanding



Problem Areas & Design

- Poor door verification practice.
- Absence of smart home security and low trust in technology.
- Emotional fear even in safe neighbourhoods.
- Lack of communication between police and citizens about preventive steps.
- Potential design space in awareness, smart alert systems, and community support tools.

Design Opportunity

- Elderly and women remain most vulnerable due to predictable lifestyles and weak safety measures.
- Burglars exploit opportunity, routine, and low awareness rather than advanced tools.

Need for Design Interventions:

- Door-interaction verification systems.
- Simple community alert tools for elderly.
- Awareness materials focused on daily preventive habits.

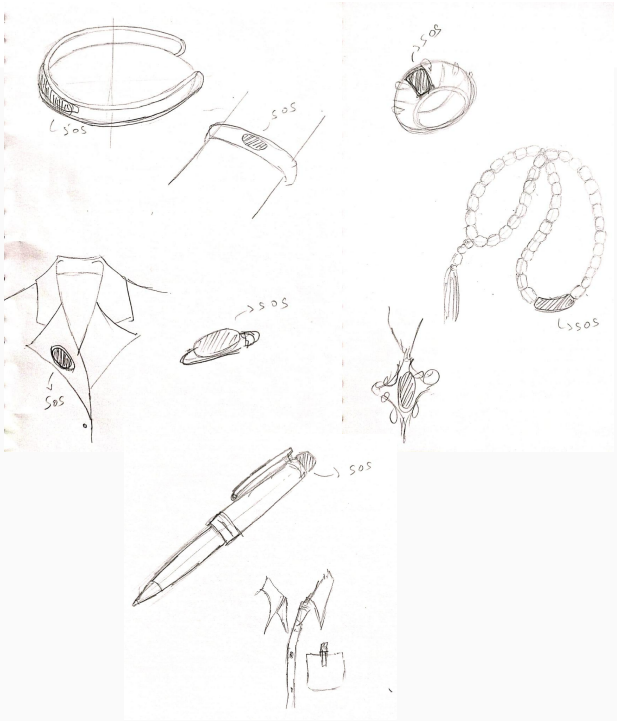
Ideation Journey

We explored multiple directions — home alert systems, door-verification tools, community support models, and wearables.

User feedback highlighted three needs:

- Always accessible
- Non-stigmatizing
- Effortless to use

These insights shifted our direction toward safety accessories.



Why Wearable Accessories?

- **Always-On & Always-Within-Reach:** Emergencies happen suddenly — falls, dizziness, panic. Unlike phones, accessories are always worn, ensuring safety is literally on the body 24×7.

- **Simple, One-Tap Use:** Most elderly users are not comfortable with smartphones or apps. A wearable accessory with just one easy press avoids menus, screens, or instructions. Even someone who is digitally non-literate can ask for help instantly.

- **No Stigma or "Medical Device" Look:** Bulky panic devices make seniors feel old, sick or monitored. Our accessory blends into daily life — it looks like jewelry, not a medical gadget, protecting both dignity and comfort.

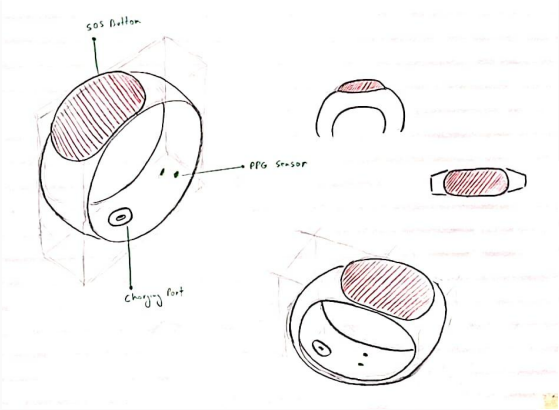
- **Culturally Familiar & Habit-Forming**: In India, rings and pendants are already part of daily routines. Elders are more likely to remember to wear something that feels natural and familiar.

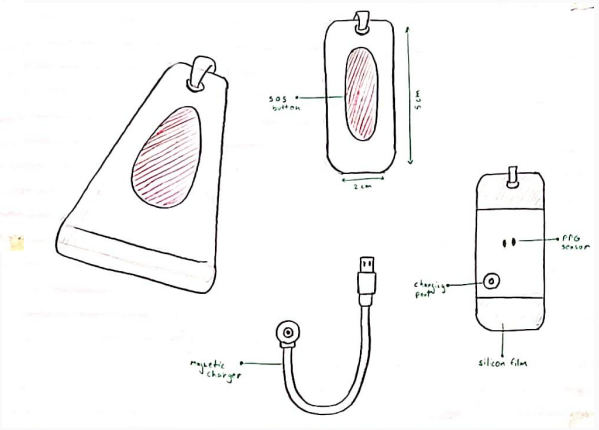
- **Hands-Free & Comfortable for Weak Grip**: Wearables eliminate dependency on fine motor skills — a tap or auto-detection is enough.

Concept Selection

Among all variations, the ring and pendant offered the best balance between:

- Accessibility
- Comfort
- Emotional acceptance
- Wearability throughout the day

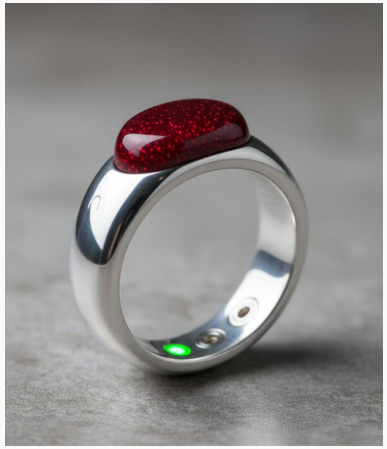




Final Product Overview

SafeTap — Smart Ring & Pendant

A discreet emergency-response accessory designed for elderly living alone. Blends **aesthetics, technology, and emotional reassurance**.







Key Features

- **SOS Button:** Instant help request sent to 3 immediate neighbours & security guard
- **PPG Sensor:** Tracks pulse irregularities and sends alerts
- **Accelerometer:** Detects falls automatically
- **Magnetic Charging:** Easy, elderly-friendly power system
- **Bluetooth Connectivity:** Syncs with app for alert notifications
- **Discreet Design:** Looks like jewelry, not medical equipment

Material

- **316L Surgical Stainless Steel:** Hypoallergenic, sweat-proof, jewelry-grade.
- **Medical-Grade Silicone Patch (Pendant Back):** Soft, skin-safe, ensures good PPG skin contact.
- **Resin-Topped SOS Button:** Lightweight, red-tinted resin that looks like a gem but is affordable.

Conclusion

Our project began by understanding how house-breaking crimes and emergencies disproportionately affect elderly people living alone. Through primary and secondary research, we discovered a deeper issue — not just lack of security, but a lack of *confidence, independence, and dignity*.

By listening to users and analysing their daily struggles, we realised that safety solutions must be simple, wearable, and emotionally acceptable. This led us to develop a smart ring and pendant that combine SOS alerts, fall detection, and pulse monitoring within a jewellery-like design.

Our solution does not aim to control elderly lives. It aims to support them quietly, giving them the reassurance that help is always within reach. It transforms safety from a burden into a gentle, elegant presence in their everyday life.